Mochi

User Story 1a - Auth & Security: As an end user I can register a new account:

1. A new account is created when all required fields are filled in correctly and the email is not already in use.
2. A clear error message is shown if a required field is missing or the email is already taken, and no account is created.

User Story 1b - Auth & Security: As an end user I can log into my account

1. A user with valid credentials is authenticated and redirected to the home screen.
2. An incorrect email or password shows the message "Invalid email or password" and the user is not logged in.

User Story 1c - Auth & Security: As an end user I can log into my account

1. Clicking the logout button invalidates the session and redirects the user to the login page.
2. After logging out, accessing a protected page redirects the user back to the login page.

User Story 2: Auth & Security: As an end user I can adjust my account settings

1. A user can update their username or email, and the change is reflected immediately with a success confirmation.

User Story 3: Scheduling & Calendar: As an end user I need a calendar to distinguish different activities

1. All scheduled activities are displayed as color-coded time blocks in the correct day and time slot.
2. A newly added activity appears immediately in the calendar in its activity-type color.

User story 4: Scheduling & Calendar: As an end user I can mark activities as completed or not

1. Clicking the completion toggle visually marks an activity as completed and saves the state immediately.
2. The completed state is preserved after reloading or closing the app.

User Story -05: Profile & Mochi: As an end user I have a profile page where I can view my information

1. The profile page displays the user's username, level, total XP, streak, and Mochi companion.
2. A user with no XP yet sees level 1 with 0 XP and an empty progress bar.

User Story -06: Profile & Mochi: As an end user I earn XP and level up based on completed activities.

1. Completing an activity or daily sub-goal adds the defined XP amount to the user's account, visible on the profile.
2. When XP reaches the next level threshold, the level increments by 1, a notification is shown, and the progress bar resets.

User Story -07: Goals & Streaks: As an end user I can create and manage global (long-term) goals

1. A new goal with a title is saved and appears in the goals list immediately.
2. Deleting a goal removes it and all its associated sub-goals from the app.

User Story -08: Goals & Streaks: As an end user I can set up to 3 daily sub-goals per global goal

1. A new daily sub-goal is saved under the correct goal for today and starts as incomplete.
2. Adding a 4th sub-goal to a goal shows an error message and the sub-goal is not created.

User Story -09: Goals & Streaks: As an end user I have a day-to-day streak based on my daily goals

1. Completing all required daily sub-goals increments the streak count by 1 at the end of the day.
2. Failing to complete the required sub-goals on a given day resets the streak to 0.

User Story -10: Scheduling & Calendar: As an end user I can define my ideal day preferences

1. A user can assign preferred activity types and set the preferred length of activities which indicates how much time does the user want to spend on the activity type within 1 day.
2. If no ideal day has been set up, the user sees a prompt with a direct link to the settings before using the schedule suggestion feature.

User Story -11: Scheduling & Calendar: As an end user the system suggests a daily schedule based on my ideal day template

1. Suggested activities are placed only in time slots that are not already occupied on today's calendar.

User Story -12: Profile & Mochi: As an end user I have a Mochi companion visible on my profile

1. The Mochi companion is displayed in a dedicated section on the user's profile page.
2. Mochi's appearance changes to match the user's current level tier when an evolution threshold is reached.

User Story 13: Social & Leaderboard: As an end user I can add friends to my account

1. When a friend request is accepted, both users appear in each other's friends lists.
2. Declining a friend request removes it without creating a connection.

User Story -14: Social & Leaderboard: As an end user I can see a leaderboard ranking me and my friends

1. The leaderboard for each challenge shows the user and all friends ranked by challenge goal completion score, with the user's entry highlighted.

User Story -15: Analytics: As an end user I can see statistics about my scheduled activities over a week

1. The statistics page shows a chart with activity completions and time spent per activity type for the past 7 days.
2. Days with no activities show as 0 — all 7 days are always visible in the chart.

User Story -16: Auth & Security: As the product owner the application is secure for end users

1. All database queries use prepared statements and all output escaped, so SQL injection and XSS attacks have no effect.
2. Any request to an authenticated endpoint without a valid session token returns HTTP 401 and no user data.

Could-Have 01: Customization: As an end user I can earn achievements based on certain criteria

1. An achievement is unlocked and shown as a badge on the profile when the user's data meets the defined criteria.
2. The user receives a visible notification when a new achievement is earned.

Could-Have 02: Customization: As an end user I can customize my Mochi through costumes or uploaded images

1. Selecting a preset costume updates Mochi's appearance on the profile.
2. Uploading a valid image replaces the default Mochi appearance with the uploaded one.

Could have 03: Customization: As an end user I can customize the background and layout of the application

1. Selecting a color scheme or background in theme settings updates the app's appearance immediately.
2. The chosen theme is still applied after logging out and back in.

Could have 04: Customization: As an end user my Mochi could evolve based on levels

1. Mochi displays the evolved appearance matching the user's level tier when an evolution threshold is reached.

Could have 05: Social & Leaderboard: As an end user I can set up challenges between friends

1. When a friend accepts a challenge, participants can see each other's progress towards the goal.
2. After the deadline, the final leaderboard is displayed to participants.